



Using a Boron-Based Anion Receptor Additive to Improve the Thermal Stability of LiPF₆-Based Electrolyte for Lithium Batteries

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The possibility of using a strong anion coordinate agent, tris(pentafluorophenyl) borane (TPFPB), to suppress the thermal decomposition of LiPF₆-based electrolyte was studied. Cyclic voltammogram measurements showed that addition of 0.1 M TPFPB maintained electrochemical stability of a LiPF₆-based electrolyte at 55°C for a week, while under the same conditions severe degradation in electrochemical stability was observed in the same LiPF₆-based electrolyte without the TPFPB additive. A Li/LiMn₂O₄ cell with a composite LiPF₆-based electrolyte containing 0.1 M TPFPB additive also exhibited superior capacity retention and cycling efficiency at 55°C than a cell with an electrolyte without additive. These data demonstrate the excellent effect of TPFPB additive in improving the thermal stability of LiPF₆-based electrolyte.

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Lithium hexafluorophosphate (LiPF₆) is by far the most commonly used salt in electrolytes of secondary lithium batteries because of its high ionic conductivity. However, LiPF₆ still suffers from both thermal and hydrolytic instability. It can undergo decomposition in solution: LiPF₆(sol) → LiF(s) + PF₅(sol).¹ The P-F bonds are also highly susceptible to hydrolysis by even trace amounts of moisture.²⁻⁴ Increased temperature accelerates decomposition. Both thermal decomposition and hydrolysis of LiPF₆ salt seriously deteriorate the performance of the electrolyte in many aspects. LiF produced from the thermal decomposition precipitates from the solution, which reduces both conductivity and numbers of free Li⁺ ions. LiF precipitation was also found at the surface of electrodes and in the pores of the separator membrane, which may block the transportation of the Li⁺ ion and increase the resistance of cells.^{5,6} Highly reactive PF₅ is produced from both the thermal and hydrolysis reactions and can initiate the polymerization of cyclic carbonate to degrade the electrolyte.¹ The instability of LiPF₆-based electrolyte is a major factor for degradation of lithium batteries operating at elevated temperatures. Although numerous salts have been synthesized, none of them can surpass LiPF₆ salt due to its superior combination of high conductivity in various organic solvents, a wide electrochemical stability window, and its ability to passivate Al and prevent current collect corrosion in the cathode.

Tasaki and Nakamura recently suggested that promotion of ion pair dissociation of LiPF₆ may prevent thermal decomposition of LiPF₆.⁷ Their molecular dynamic simulations show that the average distances between the Li⁺ cation and the F atoms in the PF₆⁻ anion in the presence of anion receptors (aza ether compounds) are much larger than the absence of the aza ether compounds. They postulated that this enlarged distance suppresses the reactivity of the Li⁺ cation and F atoms. However, most available anion receptors are based either on positively charged sites, hydrogen bonding, or Lewis acid metal centers. None of these is suitable for use in nonaqueous electrolytes. Only two families of stable anion complexing agents have been synthesized and reported by our group recently. One of these is based on aza-ether compounds.⁸ An electron withdrawing group such as CF₃SO₂⁻ was attached on the N atoms of these aza ether compounds. They function with anions in much the same way as crown ethers act with cations. Another is based on electron deficient borate or borane compounds with various fluorinated aryl or alkyl

groups.⁹ The anion coordination effect of both types of compounds has been verified by near edge X-ray absorption spectra and conductivity measurements.

However, the CF₃SO₂ substitute aza ether compounds were found to be unstable in the LiPF₆-based electrolytes. Trace of HF acid in the electrolyte could degrade the aza ether compounds. The fluorinated boron-based anion receptors display a stronger anion coordination effect than the aza ether based compounds. They can even dissolve LiF in organic solvents. Among a dozen boron-based anion receptors that we have synthesized, tris(pentafluorophenyl) borane (TPFPB) has exceptional chemical, electrochemical, and thermal stability. We have reported previously on a composite electrolyte containing LiF and TPFPB that exhibits excellent thermal stability.² We surmised that the strong anion coordinate agent TPFPB may be able to improve the thermal stability of the LiPF₆-based electrolyte. In this article we summarized our evaluation of using TPFPB to improve the stability of the LiPF₆-based electrolyte at elevated temperatures.

Experimental

TPFPB compound was synthesized as described in our previous work.⁸ The commercial electrolyte, 1 M LiPF₆ in ethyl carbonate (EC)/dimethyl carbonate (DMC) (1:1 volume ratio), was purchased from EM Industries Inc. LiMn₂O₄ spinel was obtained from FMC Inc.

Electrochemical characterization was performed in a standard coin cell configuration. Composite electrolyte was prepared by adding 0.1 M TPFPB to a commercial LiPF₆ (EC/DMC 1:1) electrolyte from EM Industry Inc. LiMn₂O₄ electrodes (typical weight 10 mg, area 2.54 cm²) were made by mixing the cathode powder with 10% acetylene black and 10% poly(vinylidene fluoride) (PVdF, KynarFlex 2801, Atochem) in 1-methyl-2-pyrrolidone, coating this slurry onto an aluminum foil and then drying the electrode in a vacuum at 100°C. A Solatron SI 1287 electrochemical interface system was used for cyclic voltammetry (CV) experiments under software control. The scan rate was 0.1 mV/s. CV was collected before and after the test cells were stored at 55°C for a week. Galvanostatic cycling was carried out on an Arbin battery test system and the cells were cycled at the C/3 rate under a constant current density of 0.12 mA/cm² at room temperature or 55°C. Elevated temperature cycling was performed with the coin cell in a vial immersed in an oil bath.

Results and Discussion

TPFPB dissolves in a LiPF₆-based (EC-DMC 1:1, v/v, EM Industry Inc.) electrolyte very quickly, while its dissolution in pure EC-DMC solvent is much slower. This confirms the interaction be-

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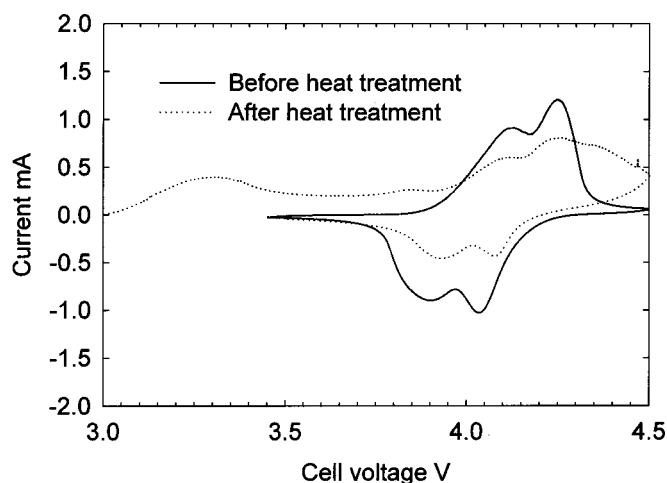


Figure 1. CV of a Li/LiMn₂O₄ cell with a commercial LiPF₆-based electrolyte before and after storage at 55°C for a week. (Sweep rate = 0.1 mV/s.)

tween LiPF₆ salt and TFPFB additive. In an attempt to ascertain the effect of the TFPFB additive on the thermal stability of the LiPF₆-based electrolyte, cyclic voltammetry was performed on a Li/LiMn₂O₄ cell with a composite LiPF₆-based electrolyte (EC/DMC 1:1) containing 0.1 M TFPFB. Data were collected from the cell before and after storage at 55°C for 7 days. Parallel measurements were also performed on a cell with a commercial LiPF₆-based electrolyte (EC/DMC 1:1, EM Industries Inc.) under the same conditions and the CVs are shown in Fig. 1. After storage at 55°C for 7 days, the performance of the cell with the commercial LiPF₆-based electrolyte shows extensive capacity fading (about 50%). An extra oxidation peak appeared at 3.25 V during the charge and the oxidation current remained at a high level up to 4.5 V. These results suggest severe degradation of the electrolyte after the cell had been stored at 55°C. Apparently, thermal decomposition of LiPF₆ has severely degraded the electrochemical stability of the electrolyte. It has been reported that PF₅ has very high reactivity with carbonate solvents.¹ We speculate that the reduced electrochemical stability is due mainly to the reaction of PF₅ with carbonate solvents. Figure 2 compares the CVs collected from the Li/LiMn₂O₄ cell with the composite LiPF₆-based electrolyte before and after heat-treatment.

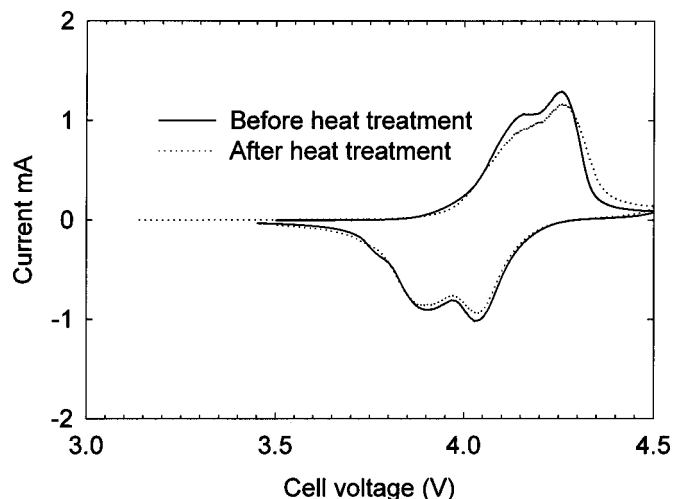


Figure 2. CV of a Li/LiMn₂O₄ cell with a composite LiPF₆-based electrolyte containing 0.1 M TFPFB additive before and after storage at 55°C for a week. (Sweep rate = 0.1 mV/s.)

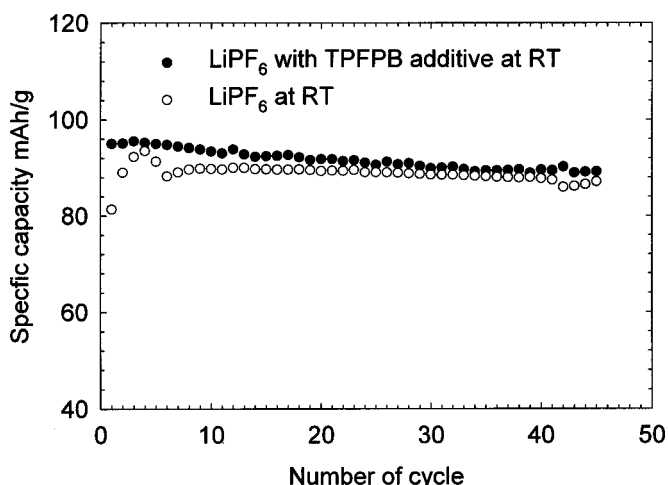


Figure 3. Comparison of the discharge capacity of Li/LiMn₂O₄ cells with a commercial LiPF₆-based electrolyte and a composite LiPF₆-based electrolyte containing 0.1 M TFPFB additive at room temperature.

The cell shows almost identical CV profiles before and after heat-treatment. The oxidation currents in the CVs, before and after heat-treatment, all approached a low background at 4.5 V which indicates excellent electrochemical stability. Furthermore, the discharge capacity dropped only by about 3% after heat-treatment, which is much lower than that from the commercial LiPF₆-based electrolyte. These observations indicate that addition of TFPFB maintained excellent electrochemical stability of LiPF₆ electrolyte at 55°C.

To further confirm the effect of the TFPFB on improving performance of the LiPF₆-based electrolyte at elevated temperatures, Li/LiMn₂O₄ cells with the commercial LiPF₆-based electrolyte and the composite electrolyte containing 0.1 M TFPFB were cycled at C/3 rate under constant current conditions. The experiments were run at both room temperature and 55°C. Figure 3 shows the capacity variation of Li/LiMn₂O₄ cells cycled at room temperature. The cell with the composite LiPF₆-based electrolyte displays 6.6% capacity loss at the 45th cycle, which is identical to that obtained from the cell with the commercial LiPF₆ electrolyte. This indicates that the presence of TFPFB additive does not have any adverse effect on either electrochemical stability of the electrolyte or compatibility with the LiMn₂O₄ cathode. Figure 4 shows coulombic efficiency (CE) of these cells. In the first few cycles, the cell with the composite electrolyte exhibited a higher CE than the cell with the commercial electrolyte. After that both cells displayed almost identical CEs. The higher initial CE and less fluctuation of CE during the cycling of the cell with the composite electrolyte imply that the TFPFB additive affects the formation of the solid electrolyte interface (SEI) layer in a favorable way.

Figure 5 shows representative charge/discharge curves of Li/LiMn₂O₄ cells cycling at 55°C. In the cell with the commercial LiPF₆-based electrolyte, in addition to severe capacity fading and polarization in discharge, the charge process became longer and longer with cycling and exhibited fluctuating voltage profiles. These results suggest some other oxidation reactions on the cathode surface besides the extraction of lithium. This was also observed in the CV scanning, as mentioned earlier, and further confirms that the thermal decomposition of LiPF₆ decreases the electrochemical stability of the electrolyte. Unlike the cell with the commercial electrolyte, the cell with the composite electrolyte exhibited negligible polarization, smooth charge/discharge profiles, and high cycling efficiency. Figure 6 compares the variation of capacity vs. cycle number obtained from cells with the commercial electrolyte and the composite electrolyte when cycling at 55°C. The cell with the composite electrolyte lost only 13% capacity in 50 cycles, while the cell

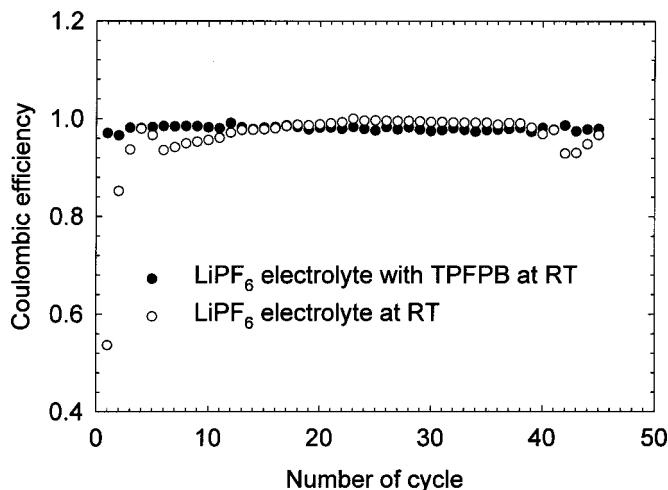


Figure 4. Comparison of the coulombic efficiency of Li/LiMn₂O₄ cells with a commercial LiPF₆-based electrolyte and a composite LiPF₆-based electrolyte containing 0.1 M TFPFB additive at room temperature.

with the commercial electrolyte lost 44% capacity. The CEs of these cells are shown in Fig. 7. The cycling efficiency of the cell with the commercial electrolyte dropped rapidly with cycling and was only 0.36 on the 50th cycle, while the cell with the composite electrolyte

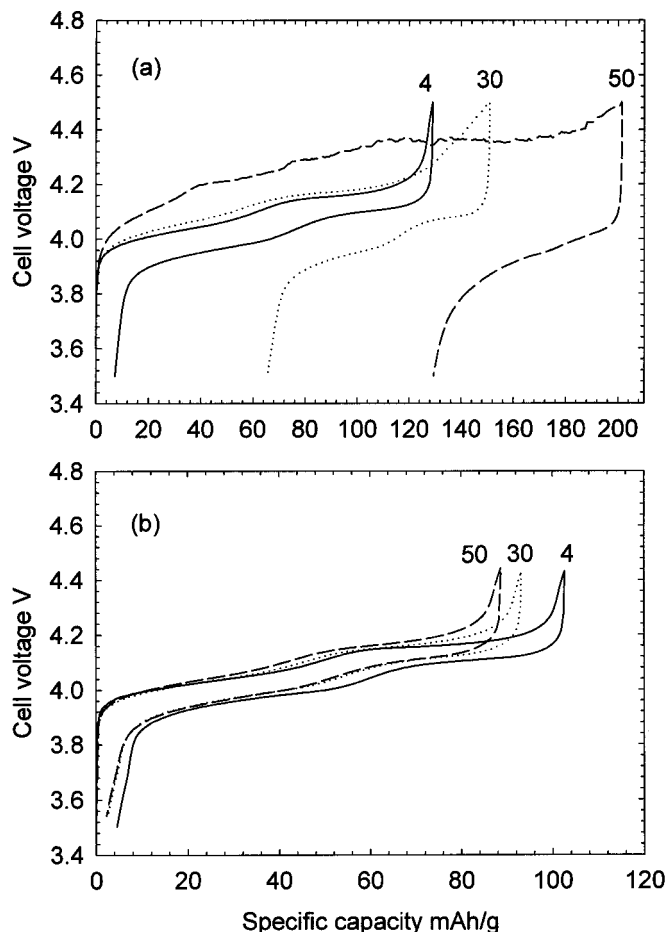


Figure 5. Voltage profiles of Li/LiMn₂O₄ cells with (a) a commercial LiPF₆-based electrolyte and (b) a composite LiPF₆-based electrolyte containing 0.1 M TFPFB additive after 4, 30, and 50 cycles at 55°C.

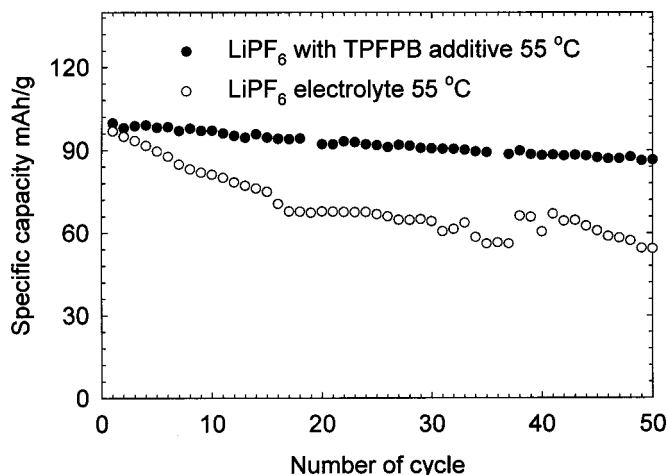


Figure 6. Comparison of the discharge capacity of Li/LiMn₂O₄ cells with a commercial LiPF₆-based electrolyte and a composite LiPF₆-based electrolyte containing 0.1 M TFPFB additive at 55°C.

maintained a very high CE (around 0.98) over 50 cycles except for the first few cycles. These results clearly show that the TFPFB additive dramatically improved thermal stability and maintained excellent electrochemical stability of the LiPF₆-based electrolyte at 55°C. We believe this effect is due to the strong anion coordination effect of TFPFB additive, which enhances Li⁺PF₆⁻ ion-pair dissociation.

Conclusions

A boron-based anion receptor, TFPFB, has been examined for suppressing the thermal decomposition of a LiPF₆-based electrolyte. CV studies show that the LiPF₆-based electrolyte experienced severe degradation in electrochemical stability after being stored at an elevated temperature 55°C for a week. The presence of 0.1 M TFPFB effectively maintained the electrochemical stability of the LiPF₆-based electrolyte under the same conditions. A Li/LiMn₂O₄ cell with a composite LiPF₆-based electrolyte containing 0.1 M TFPFB additive also exhibits much higher capacity retention and cycling efficiency at 55°C than the cell with a commercial electrolyte without the TFPFB additive. The significant improvement of the LiPF₆-based electrolyte by the TFPFB additive is believed to be due to the enhancement of Li⁺PF₆⁻ ion-pair dissociation.

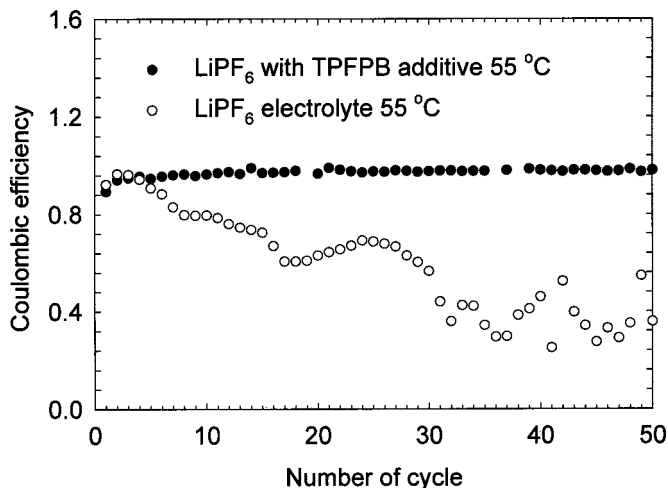


Figure 7. Comparison of the coulombic efficiency of Li/LiMn₂O₄ cells with a commercial LiPF₆-based electrolyte and a composite LiPF₆-based electrolyte containing 0.1 M TFPFB additive at 55°C.

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